MQTT Firewall Intrusion Detection Report

Generated: 2026-05-02 15:18:40
Source file: `JSONL (52 verdicts) + .log (52 verdicts)`
Time range: 15:06:19 → 15:18:30

---

1. Executive Summary

The LightGBM-based MQTT firewall processed **349 packets** (408 log lines) and issued **53 flow-level verdicts**. Of these, **49 (92.5%) were classified as attacks** and **4 (7.5%) as benign**. Mean model confidence on attack flows was **0.9621**.

Metric	Value
Total log lines	408
Total packets intercepted	349
Total flow verdicts	53
Attack verdicts	49 (92.5%)
Benign verdicts	4 (7.5%)
Observation window	15:06:19 → 15:18:30

2. Attack Type Breakdown

Attack Type	Verdict Count	Share (%)
MQTT-DDoS-Connect_Flood	15	28.3%
TCP_IP-DDoS-UDP1	12	22.6%
Recon-Port_Scan	12	22.6%
benign	4	7.5%
MQTT-DDoS-Publish_Flood	3	5.7%
MQTT-DoS-Connect_Flood	3	5.7%
ARP_Spoofing	3	5.7%
TCP_IP-DDoS-SYN	1	1.9%

3. Model Confidence Statistics

Statistics computed over **attack-classified flows only**.

Statistic	Value
Sample size	49 flows
Minimum confidence	0.6199
Maximum confidence	1.0000
Mean confidence	0.9621
Flows with confidence ≥ 0.90	39 (79.6%)

4. Flow Statistics

Metric	Min	Max	Mean
Packets per flow	3	10	6.2
Bytes per flow	0	2,900	737
Flow duration (s)	0.0005	28.1485	1.8504

5. Timing Performance — Per Attack Type

> **infer** = `model.predict()` only | **classify** = feature extraction + inference + decision | **inspect** = packet arrival → verdict (inline classifications only; 0 for background reaper-triggered flows)

Attack Type	N	Infer Mean (ms)	Infer Std (ms)	Classify Mean (ms)	Classify Std (ms)	Inspect Mean (ms)	Inspect Std (ms)
ARP_Spoofing	3	0.761	0.347	1.874	1.031	0.000	0.000
MQTT-DDoS-Connect_Flood	15	0.450	0.022	1.060	0.056	1.285	0.119
MQTT-DDoS-Publish_Flood	3	0.411	0.034	0.984	0.051	1.258	0.144
MQTT-DoS-Connect_Flood	3	0.446	0.011	1.107	0.038	0.000	0.000
Recon-Port_Scan	12	1.328	1.029	2.692	1.681	0.000	0.000
TCP_IP-DDoS-SYN	1	0.526	0.000	2.154	0.000	2.328	0.000

Attack Type	N	Infer Mean (ms)	Infer Std (ms)	Classify Mean (ms)	Classify Std (ms)	Inspect Mean (ms)	Inspect Std (ms)
TCP_IP-DDoS-UDP1	12	0.635	0.225	1.515	0.344	0.000	0.000
benign	4	0.451	0.013	1.065	0.055	1.277	0.177
Overall	53	0.708	0.616	1.598	1.072	1.325	0.251

6. Top Attacker Flows

23 distinct `(src_ip:src_port, attack_type)` flows detected.

Source (ip:port)	Classified Attack Type	Verdict Count
127.0.0.1:59169	Recon-Port_Scan	11
127.0.0.1:1883	MQTT-DDoS-Connect_Flood	10
127.0.0.1:59169	MQTT-DDoS-Connect_Flood	5
127.0.0.1:1883	MQTT-DoS-Connect_Flood	2
10.255.255.254:44717	TCP_IP-DDoS-UDP1	2
192.168.208.1:137	ARP_Spoofing	2
127.0.0.1:59169	TCP_IP-DDoS-SYN	1
10.255.255.254:46222	TCP_IP-DDoS-UDP1	1
127.0.0.1:39933	MQTT-DDoS-Publish_Flood	1
127.0.0.1:60737	MQTT-DDoS-Publish_Flood	1
10.255.255.254:59019	TCP_IP-DDoS-UDP1	1
127.0.0.1:46679	MQTT-DDoS-Publish_Flood	1
10.255.255.254:46725	TCP_IP-DDoS-UDP1	1
127.0.0.1:60737	MQTT-DoS-Connect_Flood	1
127.0.0.1:1883	Recon-Port_Scan	1
10.255.255.254:49492	TCP_IP-DDoS-UDP1	1
10.255.255.254:58290	TCP_IP-DDoS-UDP1	1
10.255.255.254:55930	TCP_IP-DDoS-UDP1	1
10.255.255.254:47186	ARP_Spoofing	1
10.255.255.254:46307	TCP_IP-DDoS-UDP1	1

...and 3 additional flows not shown.

7. Last 10 Verdicts

Time	Verdict	Attack Type	Confidence	Source	Destination	Pk
15:14:16	ATTACK	TCP_IP-DDoS-UDP1	0.9559	10.255.255.254:46307	10.255.255.254:53	4
15:14:30	ATTACK	Recon-Port_Scan	1.0000	127.0.0.1:59169	127.0.0.1:1883	3
15:14:51	ATTACK	TCP_IP-DDoS-UDP1	0.8766	10.255.255.254:58380	10.255.255.254:53	4
15:14:54	ATTACK	TCP_IP-DDoS-UDP1	0.8827	10.255.255.254:42228	10.255.255.254:53	4
15:15:30	ATTACK	Recon-Port_Scan	1.0000	127.0.0.1:59169	127.0.0.1:1883	3
15:15:54	ATTACK	TCP_IP-DDoS-UDP1	0.9523	10.255.255.254:38981	10.255.255.254:53	4
15:16:30	ATTACK	Recon-Port_Scan	1.0000	127.0.0.1:59169	127.0.0.1:1883	3
15:17:30	ATTACK	Recon-Port_Scan	1.0000	127.0.0.1:59169	127.0.0.1:1883	3
15:17:39	ATTACK	ARP_Spoofing	0.8956	192.168.208.1:137	192.168.223.255:137	3
15:18:30	ATTACK	Recon-Port_Scan	1.0000	127.0.0.1:59169	127.0.0.1:1883	3

Report generated by `scripts/parse_firewall_logs.py` — MQTT IDS/IPS Benchmarking Framework.